

Coordination failure

These theories often highlight the problem that several things must work well enough, at the same time, to get sustainable development under way.

Complementarity: An action taken by one firm, worker, or organization that increases the incentives for other agents to take similar actions.

Coordination failure is a state of affairs in which agents' inability to coordinate their behavior (choices) leads to an outcome (equilibrium) that leaves all agents worse off than in an alternative situation that is also an equilibrium.

They simply cannot get there because of difficulties of coordination, sometimes because people hold **different expectations** and sometimes because **everyone is better off waiting** for someone else to make the first move.

Examples:

1. presence of firms using specialized skills and the availability of workers who have acquired those skills
2. commercialization of agriculture.
3. "chicken and egg" problem
4. complementary investments must come at the same time, through coordination
5. where-to-meet dilemma

Multiple equilibria

A condition in which more than one equilibrium exists. These equilibria may sometimes be ranked, in the sense that one is preferred to another, but the unaided market will not move the economy to the preferred outcome.

The benefits an agent receives from taking an action depend positively on how many other agents are expected to take the action

For example, the price a farmer can hope to receive for his produce

Multiple equilibria

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Equilibrium is found where the “privately rational decision function” crosses the 45-degree line.

“agents observe what they expected to observe”

The process of adjustment of expectations would continue until the function crosses the 45° line.

level of actual investment = level of expected investment

D1 and D3 are “stable” equilibria in each of these two, the S-shaped function cuts the 45-degree line from above

At the middle equilibrium at D2, the function cuts the 45-degree line from below, and so it is unstable.

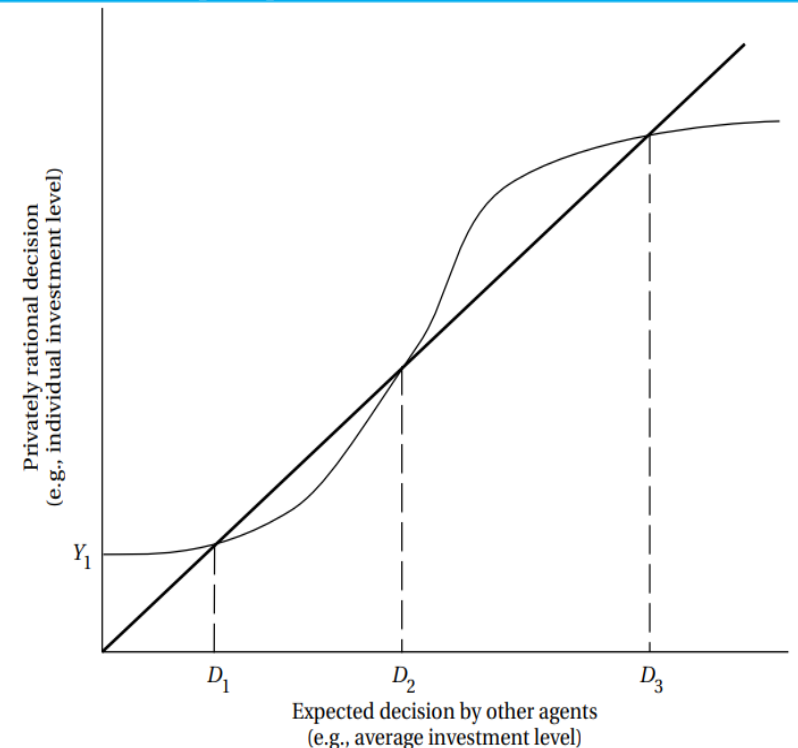
All are better off with more investors or higher rates of investment, but the market may not get us there without the influence of certain types of government policy

Market forces can generally bring us to one of the equilibria, but they are not sufficient to ensure that the best equilibrium will be achieved, and they offer no mechanism to become unstuck from a bad equilibrium and move toward a better one

The economy can not move to the higher equilibrium due to coordination failure among the firms

The economy can not develop due to coordination failure

FIGURE 4.1 Multiple Equilibria



Big push

A “big push” or a large comprehensive program is needed in the form of a high minimum amount of investment to overcome the obstacles to development

“bit by bit” investment will not launch the economy successfully on the development path,

A critical ground speed which must be passed before the aircraft can become airborne

It necessitates the obtaining of external economies that arise from the simultaneous establishment of technically interdependent industries

indivisibilities and external economies flowing from a minimum quantum of investment are a prerequisite

1. indivisibilities in supply of social overhead capita
2. indivisibility of demand;
3. indivisibility in the supply of savings

Indivisibilities in the production function

basic industries like power, transport, and communications are indirectly productive and have a long gestation period. They cannot be imported. Their installations require a “sizeable initial lump” of investment 30-40 per cent of its total investment in these channels

Indivisibility of demand

The indivisibility of demand requires simultaneous setting up of interdependent industries in underdeveloped countries because low incomes limit the demand for their products necessitates a high minimum quantum of investment in interdependent industries to enlarge the size of the market.

Indivisibility of supply of savings

A high minimum size of investment requires a high volume of savings marginal rate of saving should be very much higher

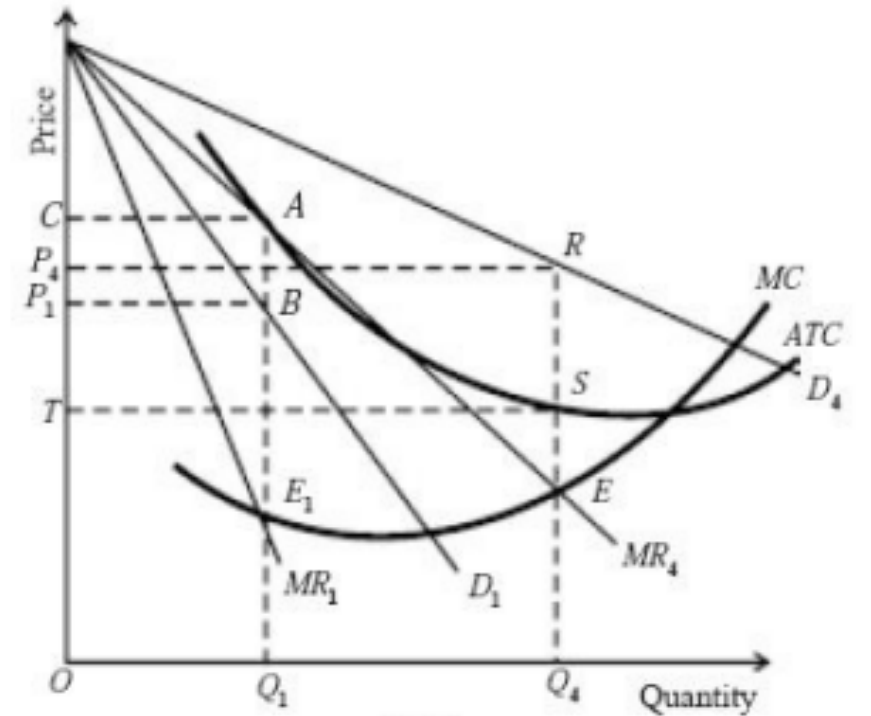


Fig. 1



Michael Kremer's 0 ring theory

It explains not only the existence of poverty traps but also the reasons that countries caught in such traps

It models production with strong **complementarities among inputs**.

Suppose that a production process is broken down into n tasks. There are many ways of carrying out these tasks, which for simplicity we order strictly by level of skill q required, where q is between zero and one. The higher the skill, the higher the probability that the task will be “successfully completed”

Output is given by multiplying the q values of each of the n tasks together, in turn multiplied by B , that depends on the characteristics of the firm and here its value is 1. Suppose also that each firm hires only two workers. Then the O-ring production function looks like this

$$BF(q_i, q_j) = q_i q_j$$

One of the most prominent features of this type of production function is what is termed **positive assortative matching**. This means that workers with high skills will work together and workers with low skills will work together.

this type of matching means that high-value products will be concentrated in countries with high-value skills.

everyone would like to work with the more productive workers

you will be more productive when working with a more productive person

there will be a strong tendency for the most productive workers to work together

Total output will always be higher under a matching scheme

$$q_H^2 + q_L^2 > 2q_H q_L$$

Examples: a symphony orchestra, the best jazz performers play and record together, The restaurant with the very best chef also hires mature, highly trained, full-time waiters, “marriage market”

some firms and workers, even an entire low-income economy, can fall into a **trap of low skill and low productivity**, while others escape into higher productivity

Implications

1. Firms tend to employ workers with similar skills
2. Workers performing the same task earn higher wages in a high-skill firm than in a low-skill firm
3. Because wages increase in q at an increasing rate, wages will be more than proportionally higher in developed countries
4. when those around you have higher average skills, you have a greater incentive to acquire more skills
5. To come out of the trap, the economies will have to **enhance the productivity** of their worker and also depend on **positive assortative matching**.
6. The difference in the incomes of the countries is due to difference in productivities.
7. Human capital formation enhances productivity of workers
8. Firms expand due to higher productivity and higher profits